Last (Family) name: _____

First (Given) name: _____

Student ID (10 digits): _____

University of Toronto
Faculty of Arts & Science

DECEMBER 2022 EXAMINATIONS

STA314H1F
Statistical Methods for Machine Learning I

Duration: 2 hours
Aids Allowed: None

Exam Reminders:

- Fill out your name and student number on the top of this page.
- If a scantron and/or exam booklets are required: Ensure you fill in your name and student number on the scantron and/or exam booklet(s)
- Do not begin writing the actual exam until the announcements have ended and the Exam Facilitator has started the exam.
- As a student, you help create a fair and inclusive writing environment. If you possess an unauthorized aid during an exam, you may be charged with an academic offence.
- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. Do not collect your bag and jacket before your exam is handed in.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.
- In the event of a fire alarm, do not check your cell phone when escorted outside.

Special Instructions: We have 6 problems in total. Answer Problems 1 - 5 on the exam paper. Answer Problem 6, the multiple choices questions, on the scantron sheet.

Exam Format and Grading Scheme: The exam has 40 points in total. Each sub-question has its points indicated.

Students must hand in all examination materials at the end

Problem 1 (3 points)

Suppose we have a data set which only contains two samples $y_1 = 2$ and $y_2 = 4$. We will apply the bootstrap method to estimate the standard error of the sample mean of y_1 and y_2.

(a) (2 points) Our first step is to create some bootstrap samples. Write down **all possible different** bootstrap samples. Note: we treat $(y_1, y_2) = (2, 4)$ and $(y_1, y_2) = (4, 2)$ as different samples.

(b) (1 point) In practice, we will use R to perform boostrap. Based on the following output of R, what is the estimated standard error of the sample mean of y_1 and y_2 by using bootstrap?

```
> library(boot)
> data_y = c(2, 4)
> mean.fn = function(data, index){
               return(mean(data[index]))
            }
> boot(data_y, mean.fn, 1000)
ORDINARY NONPARAMETRIC BOOTSTRAP

Bootstrap Statistics :
       original    bias     std. error
t1*        3      -0.004      0.69
```

Problem 2 (7 points)

Consider the problem of classifying a random, discrete, class label $Y \in \{0, 1\}$ based on a 1-dimensional feature $X \in \mathbb{R}$. We assume

$$\mathbb{P}(Y = 1) = \eta, \qquad \mathbb{P}(Y = 0) = 1 - \eta, \tag{1}$$

for some $0 < \eta < 1$, and

$$X \mid Y = k \sim N(\mu_k, 1), \qquad \forall\, k \in \{0, 1\}. \tag{2}$$

Let $(y_1, x_1), \ldots, (y_n, x_n)$ be n i.i.d. realizations of (Y, X) following (1) and (2). Answer the following questions. <u>Include intermediate steps in your answer</u>.

Recall that the probability density function of $N(\mu, \sigma^2)$ at x is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

(a) (3 points) Assume $0 < \sum_{i=1}^{n} y_i < n$. Prove that the maximum likelihood estimators (MLE) of η, μ_0 and μ_1 are

$$\widehat{\eta} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\mu}_0 = \frac{1}{n_0}\sum_{i=1}^{n}(1-y_i)x_i, \qquad \widehat{\mu}_1 = \frac{1}{n_1}\sum_{i=1}^{n} y_i x_i. \tag{3}$$

where

$$n_0 = \sum_{i=1}^{n}(1-y_i), \qquad n_1 = \sum_{i=1}^{n} y_i.$$

(Continue your answer of part (a).)

(b) (2 points) Write down the decision boundary of the Bayes classifier.

(c) (2 point) Suppose the training data set contains: $(y_1, x_1) = (1, 1)$, $(y_2, x_2) = (1, 0)$, $(y_3, x_3) = (1, 1/2)$ and $(y_4, x_4) = (0, -1/2)$. Suppose we estimate the Bayes classifier by $\widehat{f}$, the plug-in estimator based on the MLEs of η, μ_0 and μ_1 in (3).

Compute the predicted label of $\widehat{f}$ at the test point $x = -1$. (You might use the fact that $\log_e(3) > 1$)

Problem 3 (8 points)

Consider a classification problem for (Y, X), where Y is a binary random variable taking values in $\{0, 1\}$. We assume that the binary random variable Y given X is generated in the following way.

- $\varepsilon \sim \text{Uniform}[0, 1]$, that is, ε follows the uniform distribution on $[0, 1]$;
- ε is independent of X;
- $Y = 1\{2X - \varepsilon > 0\}$, where $1\{\cdot\}$ is the indicator function.

(Hint: the cumulative density function of ε is $\mathbb{P}(\varepsilon < x) = x$ for any $x \in [0, 1]$.)

(a) (2 points) Derive the Bayes classifier at $X = x$ for any $x \in (0, \frac{1}{2})$.

(b) (2 points) Derive the Bayes error rate at $X = x$ for any $x \in (0, \frac{1}{2})$.

(c) (2 point) Assume that we have another classifier:

$$\widehat{f}(x) = \begin{cases} 1, & \text{if } \frac{1}{3} \leq x < \frac{1}{2}; \\ 0 & \text{if } 0 < x < \frac{1}{3}. \end{cases} .$$

Suppose we have a training set with 3 data points $(y_1, x_1) = (0, \frac{1}{5})$, $(y_2, x_2) = (0, \frac{1}{3})$ and $(y_3, x_3) = (1, \frac{2}{5})$. Compute the training error rates of both the Bayes classifier and the classifier $\widehat{f}$.

(d) (2 points) Compute the expected test error rate of $\widehat{f}$ in part (c) at $X = x$ for any test point $x \in (0, \frac{1}{2})$ and compare it with the Bayes error rate at $X = x$ in part (b).

Problem 4 (6 points)

In `Carseats` dataset, we use the decision-tree approach to predict the response $Y \in \mathbb{R}$ based on a three continuous features $X = (X_1, X_2, X_3) \in \mathbb{R}^3$. The following table shows the corresponding measures of Y, X_1, X_2, X_3.

Y	`Sales`
X_1	`Advertising`
X_2	`Income`
X_3	`CompPrice`

The following figure depicts a fitted regression decision-tree by using recursive binary splitting based on the training data $(y_1, x_1), \ldots, (y_n, x_n)$ of (Y, X).

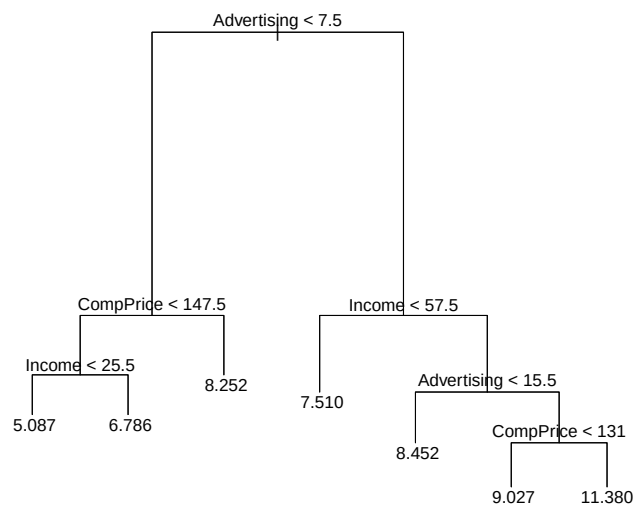


Figure 0.1: The fitted decision-tree

(a) Based on Figure 0.1, answer the following questions.

(a1) (1 point) How many times did the fitted decision-tree split the feature space?

(a2) (1 point) How many (non-overlapping) regions of the feature space did the fitted decision-tree yield?

(a3) (1 point) For any fitted decision-tree with k splits (here k is an positive integer) by using recursive binary splitting, how many regions of the feature space did the fitted decision-tree yield?

(b) For each feature X_j, $j \in \{1, 2, 3\}$, with a cutoff point $s \in \mathbb{R}$, recall that $R_1(j, s)$ and $R_2(j, s)$ form a partition of the feature space, that is,

$$R_1(j, s) := \left\{ X \in \mathbb{R}^3 \mid X_j < s \right\}, \qquad R_2(j, s) := \left\{ X \in \mathbb{R}^3 \mid X_j \geq s \right\}.$$

We define the following criterion for each feature X_j:

$$L_j := \min_{s \in \mathbb{R}} \left\{ \sum_{i : x_i \in R_1(j,s)} (y_i - \bar{y}_1)^2 + \sum_{i : x_i \in R_2(j,s)} (y_i - \bar{y}_2)^2 \right\}$$

where $\bar{y}_1$ and $\bar{y}_2$ are the averaged responses whose corresponding features are in $R_1(j, s)$ and $R_2(j, s)$, respectively.

For two features $j, k \in \{1, 2, 3\}$, X_j is said to be more important than X_k for predicting Y if $L_j \leq L_k$. Based on Figure 0.1, answer the following questions.

(b1) (1 point) Could you compare the importance of X_1 and X_2, or there isn't enough information to tell?

(b2) (1 point) Could you compare the importance of X_1 and X_3, or there isn't enough information to tell?

(b3) (1 point) Could you compare the importance of X_2 and X_3, or there isn't enough information to tell?

Problem 5 (6 points)

(a) (3 points) Consider the ridge regression with tuning parameter λ. Draw a picture which contains three curves (squared bias, variance, test MSE) of the fitted model. Use the x-axis to indicate values of λ which start from 0 and increase until a large number. The y-axis should represent the values of squared bias, variance and test MSE of the fitted models.

In your picture, indicate which curves correspond to the squared bias, variance and test MSE, respectively. Pay attention to the trend and relative magnitudes of the three metrics.

(b) Consider the estimator which solves the following problem

$$\min_{\beta=(\beta_1,\beta_2)} \sum_{i=1}^{n} (y_i - \beta_1 x_{i1} - \beta_2 x_{i2})^2, \quad \text{subject to} \quad |\beta_1| + |\beta_2| \leq s. \tag{4}$$

(b1) (1 point) In general, can the solution of (4) give us a sparse model?

(b2) (2 points) Suppose the least squares estimate of (β_1, β_2) in this example is $(\widehat{\beta}_1^{LS}, \widehat{\beta}_2^{LS}) = (-1, 1)$. If the solution to (4) is unique for $s = 2$, does it contain zeros? Please briefly explain your reasoning.

Problem 6 (10 points, 1 point for each subquestion)

Be sure to mark your answers on the answer sheet of multiple choice questions. There can be **one to four** correct answers to each question. One point is assigned to a multiple choice question **if and only if** all correct answers to this question are checked and no incorrect answer to this question is checked.

1. Which of the following statements are true

   A The Bayes classifier usually has the smallest training error rate among all possible classifiers.

   B The Bayes classifier usually has the smallest expected test error rate among the class of possible classifiers.

   C When Y has three categories (e.g., $Y \in \{1, 2, 3\}$), the Bayes error rate can not be greater than 0.5.

   D The training error rate of the Bayes classifier is always greater than 0.

2. Which of the following statements are true

   A Logistic regression is a classification method.

   B The parameters in logistic regression is estimated by the least squares approach.

   C The Bayes classifier in logistic regression has a linear decision boundary of the feature.

   D Logistic regression cannot include qualitative features.

3. Which of the following statements are true

   A LDA assumes the density function of X in each response class is Gaussian (or multivariate Gaussian).

   B The decision boundary of the Bayes classifier in LDA is linear in X.

   C LDA cannot be used when the response Y has three categories (e.g., $Y \in \{1, 2, 3\}$).

   D In LDA, we cannot estimate the conditional probability $\mathbb{P}(Y = 1 \mid X = x)$.

4. Which of the following statements are true

   A QDA is a more flexible method than LDA.

   B Naive Bayes classifier is a more flexible method than QDA.

   C QDA usually has smaller training error rate than LDA.

   D QDA usually has smaller test error rate than LDA.

5. Which of the following statements are true

   A SVM is a linear classifier, that is, its decision boundary is linear in the feature.

   B SVM cannot estimate the conditional probability $\mathbb{P}(Y = 1 \mid X = x)$.

   C SVM can only be used for linear separable data.

   D The decision boundary of SVM always depends on all the training data.

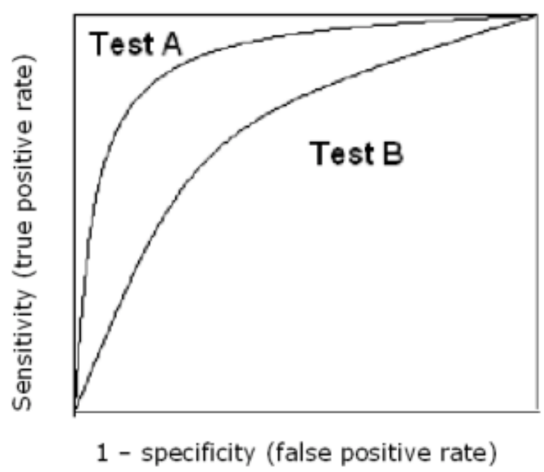
6. In the following plot, we compare two classification methods (called Test A and Test B) based on their ROC curves on the training data. Which of the following statements are true?

   A Test A is better than B in the training data set.

   B Test B is better than A in the training data set.

   C Test A has smaller test error than B.

   D Test B has smaller test error than A.



7. Which of the following statements are true

   A Bagging and random forests are used to reduce the bias of simple decision-trees.

   B Bagging and random forests are used to reduce the variance of simple decision-trees.

   C Bagging and boosting can only be used in the context of decision-trees.

   D Both bagging decision trees and random forests use bootstrap samples.

8. Which of the following statements about generalized additive models (GAMs) are true?

   A GAMs suffer from the curse of dimensionality.

   B GAMs cannot be used when we have more than 1 feature.

   C GAMs can be used in classification problems.

   D GAMs do not account for interaction of features.

9. Which of the following statements about local regression are true?

   A A larger size of the neighborhood usually yields a fitted model with smaller variance.

   B A larger size of the neighborhood usually yields a fitted model with smaller bias.

   C A larger size of the neighborhood usually yields a fitted model with smaller expected test MSE.

   D Local regression method suffers from curse of dimensionality.

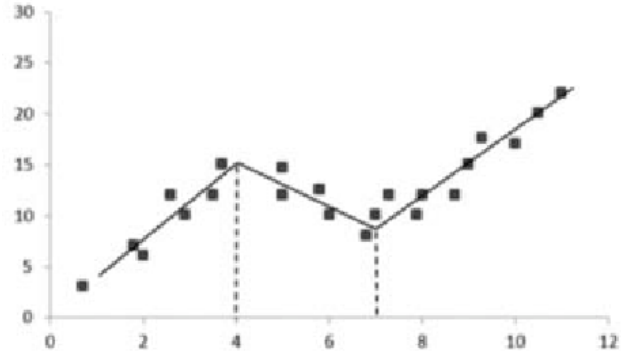
10. In the following plot, what approach did we most likely use to fit the data?

    A Polynomial regression

    B Step function approach

    C Linear spline

    D Local regression



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